

1. To get the number of customers.

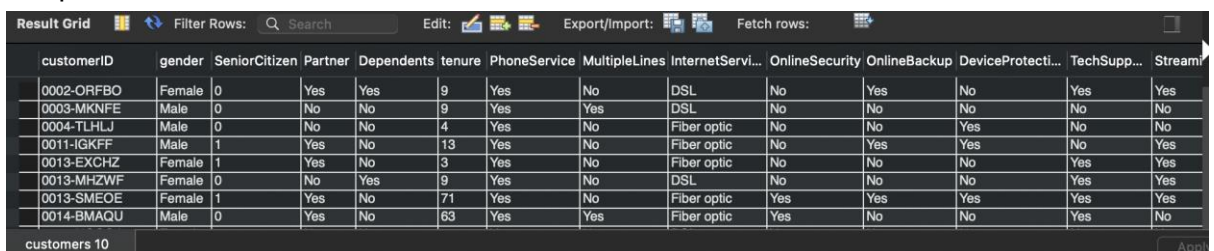
```
SELECT COUNT(*) AS total_rows FROM customers;
```

Output: 7043

2. To view the first 10 customers.

```
SELECT *  
FROM customers  
LIMIT 10;
```

Output:



The screenshot shows a database interface with a 'Result Grid' displaying the first 10 rows of the 'customers' table. The columns are: customerID, gender, SeniorCitizen, Partner, Dependents, tenure, PhoneService, MultipleLines, InternetService..., OnlineSecurity, OnlineBackup, DeviceProtecti..., TechSupp..., and Streami... The data is as follows:

customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetServi...	OnlineSecurity	OnlineBackup	DeviceProtecti...	TechSupp...	Streami...
0002-ORFBO	Female	0	Yes	Yes	9	Yes	No	DSL	No	Yes	No	Yes	Yes
0003-MKNFE	Male	0	No	No	9	Yes	Yes	DSL	No	No	No	No	No
0004-TLHLJ	Male	0	No	No	4	Yes	No	Fiber optic	No	No	Yes	No	No
0011-IGKFF	Male	1	Yes	No	13	Yes	No	Fiber optic	No	Yes	Yes	No	Yes
0013-EXCHZ	Female	1	Yes	No	3	Yes	No	Fiber optic	No	No	No	Yes	Yes
0013-MHZWF	Female	0	No	Yes	9	Yes	No	DSL	No	No	No	Yes	Yes
0013-SMEOE	Female	1	Yes	No	71	Yes	No	Fiber optic	Yes	Yes	Yes	Yes	Yes
0014-BMAQU	Male	0	Yes	No	63	Yes	Yes	Fiber optic	Yes	No	No	Yes	No

3. To check for missing values.

```
SELECT  
    SUM(CASE WHEN customerID IS NULL THEN 1 ELSE 0 END) AS missing_id,  
    SUM(CASE WHEN gender IS NULL THEN 1 ELSE 0 END) AS  
missing_gender,  
    SUM(CASE WHEN SeniorCitizen IS NULL THEN 1 ELSE 0 END) AS  
missing_senior_citizen,  
    SUM(CASE WHEN Partner IS NULL THEN 1 ELSE 0 END) AS  
missing_partner,  
    SUM(CASE WHEN Dependents IS NULL THEN 1 ELSE 0 END) AS  
missing_dependents,  
    SUM(CASE WHEN tenure IS NULL THEN 1 ELSE 0 END) AS  
missing_tenure,  
    SUM(CASE WHEN PhoneService IS NULL THEN 1 ELSE 0 END) AS  
missing_phone_service,  
    SUM(CASE WHEN InternetService IS NULL THEN 1 ELSE 0 END) AS  
missing_internet_service,  
    SUM(CASE WHEN Contract IS NULL THEN 1 ELSE 0 END) AS  
missing_contract,  
    SUM(CASE WHEN MonthlyCharges IS NULL THEN 1 ELSE 0 END) AS  
missing_monthly_charges,  
    SUM(CASE WHEN TotalCharges IS NULL THEN 1 ELSE 0 END) AS  
missing_total_charges,  
    SUM(CASE WHEN Churn IS NULL THEN 1 ELSE 0 END) AS missing_churn
```

FROM customers;

Output:

missing_id	missing_gend...	missing_senior_citiz...	missing_partn...	missing_depende...	missing_tenu...	missing_phone_servi...	missing_internet_serv...	missing_contr...	missing_montl...
0	0	0	0	0	0	0	0	0	0

4. To check for duplicate customerID's

```
SELECT
    customerID,
    COUNT(*) AS occurrences
FROM customers
GROUP BY customerID
HAVING COUNT(*) > 1;
```

Output:

customerID	occurrences

5. To check for invallied tenures. I.e where tenure lenght is less than zero.

```
SELECT COUNT(*) AS invalid_tenure
FROM customers
WHERE tenure < 0;
```

Output:

invalid_tenure
0

6. To check for invalid monthly charge. I.e where the value of monthly charge is less than or equal to zero.

```
SELECT COUNT(*) AS invalid_monthly_charges
FROM customers
WHERE MonthlyCharges <= 0;
```

Output:

invalid_monthly_charg...
0

7. To account for new customers without values in total charges yet.

```
ALTER TABLE customers
ADD COLUMN TotalCharges_clean DECIMAL(10,2);
```

```
UPDATE customers
SET TotalCharges_clean = CAST(NULLIF(TRIM(TotalCharges), '') AS
DECIMAL(10,2));
```

```
SELECT COUNT(*) AS null_total_charges
FROM customers
WHERE TotalCharges_clean IS NULL;
```

Output:

null_total_charg...
11

8. To calculate the percentage churn

```
SELECT
Churn, COUNT() AS total_customers, ROUND(COUNT() / 7043 * 100, 2) AS
percentage FROM customers GROUP BY Churn;
```

Output:

Churn	total_customers	percentage
No	5174	73.46
Yes	1869	26.54

9. What percentage of our customer base has churned?

```
SELECT COUNT() AS total_customers, SUM(CASE WHEN Churn = 'Yes' THEN 1
ELSE 0 END) AS total_churned, SUM(CASE WHEN Churn = 'No' THEN 1 ELSE 0
END) AS total_retained, ROUND(SUM(CASE WHEN Churn = 'Yes' THEN 1 ELSE 0
END) / COUNT() * 100, 2) AS churn_rate_pct FROM customers;
```

Output:

total_customers	total_churn...	total_retain...	churn_rate_pct
7043	1869	5174	26.54

10. Churn risk by contract type

```
SELECT Contract AS contract_type, COUNT() AS total_customers, SUM(CASE
WHEN Churn = 'Yes' THEN 1 ELSE 0 END) AS churned_customers, SUM(CASE
WHEN Churn = 'No' THEN 1 ELSE 0 END) AS retained_customers,
ROUND(SUM(CASE WHEN Churn = 'Yes' THEN 1 ELSE 0 END) / COUNT() * 100,
```

2) AS churn_rate_pct FROM customers GROUP BY Contract ORDER BY churn_rate_pct DESC;

Output:

	contract_type	total_customers	churned_customers	retained_custom...	churn_rate_pct
	Month-to-month	3875	1655	2220	42.71
	One year	1473	166	1307	11.27
	Two year	1695	48	1647	2.83

11. Group customers into tenure bands to find the danger zone. numbered labels above keep bands in the right order

SELECT CASE WHEN tenure <= 6 THEN '1. 0-6 months' WHEN tenure <= 12 THEN '2. 7-12 months' WHEN tenure <= 24 THEN '3. 13-24 months' WHEN tenure <= 48 THEN '4. 25-48 months' ELSE '5. 48+ months' END AS tenure_band,

COUNT(*) AS total_customers,
SUM(CASE WHEN Churn = 'Yes' THEN 1 ELSE 0 END) AS
churned_customers,
ROUND(SUM(CASE WHEN Churn = 'Yes' THEN 1 ELSE 0 END)
/ COUNT(*) * 100, 2) AS churn_rate_pct

FROM customers GROUP BY tenure_band ORDER BY tenure_band ASC;

Output:

	tenure_band	total_customers	churned_customers	churn_rate_pct
	1. 0-6 months	1481	784	52.94
	2. 7-12 months	705	253	35.89
	3. 13-24 months	1024	294	28.71
	4. 25-48 months	1594	325	20.39
	5. 48+ months	2239	213	9.51

12. Churn rate by service type

SELECT InternetService AS internet_service, COUNT() AS total_customers,
SUM(CASE WHEN Churn = 'Yes' THEN 1 ELSE 0 END) AS churned_customers,
ROUND(SUM(CASE WHEN Churn = 'Yes' THEN 1 ELSE 0 END) / COUNT() * 100,
2) AS churn_rate_pct FROM customers GROUP BY InternetService ORDER BY
churn_rate_pct DESC;

Output:

	internet_servi...	total_customers	churned_customers	churn_rate_pct
	Fiber optic	3096	1297	41.89
	DSL	2421	459	18.96
	No	1526	113	7.40

13. Financial impact. Revenue at risk by contract type.

```

SELECT
    Contract
    AS contract_type,
    SUM(CASE WHEN Churn = 'Yes' THEN 1 ELSE 0 END)
    AS
churned_customers,
    ROUND(AVG(MonthlyCharges), 2)
    AS avg_monthly_charge,

    -- Monthly revenue already lost from churned customers
    ROUND(SUM(CASE WHEN Churn = 'Yes'
        THEN MonthlyCharges ELSE 0 END), 2)
    AS
monthly_revenue_lost,

    -- Annualise it to show the full yearly impact
    ROUND(SUM(CASE WHEN Churn = 'Yes'
        THEN MonthlyCharges ELSE 0 END) * 12, 2)
    AS
annual_revenue_lost
FROM customers
GROUP BY Contract
ORDER BY annual_revenue_lost DESC;

```

Output:

contract_type	churned_customers	avg_monthly_char...	monthly_revenue_l...	annual_revenue_lo...
Month-to-month	1655	66.40	120847.10	1450165.20
One year	166	65.05	14118.45	169421.40
Two year	48	60.77	4165.30	49983.60

14. Grand total revenue at risk across all contract types

```

SELECT
    ROUND(SUM(CASE WHEN Churn = 'Yes'
        THEN MonthlyCharges ELSE 0 END), 2)
    AS total_monthly_lost,
    ROUND(SUM(CASE WHEN Churn = 'Yes'
        THEN MonthlyCharges ELSE 0 END) * 12, 2)
    AS total_annual_lost
FROM customers;

```

Output:

total_monthly_lost	total_annual_lost
139130.85	1669570.20

15. Number of additional services customers have , churn rate

```
SELECT (CASE WHEN OnlineSecurity = 'Yes' THEN 1 ELSE 0 END + CASE WHEN
OnlineBackup = 'Yes' THEN 1 ELSE 0 END + CASE WHEN DeviceProtection =
'Yes' THEN 1 ELSE 0 END + CASE WHEN TechSupport = 'Yes' THEN 1 ELSE 0 END
+ CASE WHEN StreamingTV = 'Yes' THEN 1 ELSE 0 END + CASE WHEN
StreamingMovies = 'Yes' THEN 1 ELSE 0 END) AS num_services,

COUNT(*)                               AS total_customers,
SUM(CASE WHEN Churn = 'Yes' THEN 1 ELSE 0 END)      AS
churned_customers,
ROUND(SUM(CASE WHEN Churn = 'Yes' THEN 1 ELSE 0 END)
/ COUNT(*) * 100, 2)                      AS churn_rate_pct
```

FROM customers GROUP BY num_services ORDER BY num_services ASC;

Output:

	num_services	total_customers	churned_customers	churn_rate_pct
	0	2219	475	21.41
	1	966	442	45.76
	2	1033	370	35.82
	3	1118	306	27.37
	4	852	190	22.30
	5	571	71	12.43
	6	284	15	5.28

16. Churn rate by payment method

```
SELECT PaymentMethod, COUNT() AS total_customers, SUM(CASE WHEN
Churn = 'Yes' THEN 1 ELSE 0 END) AS churned_customers, ROUND(SUM(CASE
WHEN Churn = 'Yes' THEN 1 ELSE 0 END) / COUNT() * 100, 2) AS churn_rate_pct
FROM customers GROUP BY PaymentMethod ORDER BY churn_rate_pct DESC;
```

Output:

	PaymentMethod	total_customers	churned_customers	churn_rate_pct
	Electronic check	2365	1071	45.29
	Mailed check	1612	308	19.11
	Bank transfer (automatic)	1544	258	16.71
	Credit card (automatic)	1522	232	15.24

17. Customers with high risk signals

```

SELECT
    customerID,
    Contract,
    tenure,
    InternetService,
    MonthlyCharges,
    PaymentMethod,

    -- Count how many risk factors each customer has
    (CASE WHEN Contract = 'Month-to-month' THEN 1 ELSE 0 END +
    CASE WHEN tenure <= 6 THEN 1 ELSE 0 END +
    CASE WHEN InternetService = 'Fiber optic' THEN 1 ELSE 0 END +
    CASE WHEN PaperlessBilling = 'Yes' THEN 1 ELSE 0 END +
    CASE WHEN OnlineSecurity = 'No' THEN 1 ELSE 0 END) AS risk_score

FROM customers
WHERE Churn = 'No' -- active customers only
    AND Contract = 'Month-to-month' -- highest risk contract type
    AND tenure <= 6 -- in the early danger window
ORDER BY risk_score DESC, MonthlyCharges DESC
LIMIT 50;

```

Output:

customerID	Contract	tenure	InternetServi...	MonthlyCharges	PaymentMethod	risk_score
1393-IMKZG	Month-to-month	1	Fiber optic	95.85	Electronic check	5
4929-XIHVW	Month-to-month	2	Fiber optic	95.50	Credit card (automatic)	5
5150-ITWWB	Month-to-month	3	Fiber optic	94.85	Electronic check	5
6350-XFYGW	Month-to-month	4	Fiber optic	94.75	Electronic check	5
6137-NICCO	Month-to-month	6	Fiber optic	94.05	Mailed check	5
3318-NMQXL	Month-to-month	3	Fiber optic	92.00	Electronic check	5
4927-WWOOZ	Month-to-month	2	Fiber optic	91.45	Electronic check	5
8622-ZLFKO	Month-to-month	6	Fiber optic	90.75	Electronic check	5
8266-VBFQL	Month-to-month	4	Fiber optic	90.40	Electronic check	5

18. Contract type by annual revenue lost and annual revenue retained

```

SELECT * FROM customers;
SELECT Contract, COUNT() AS total_customers,
SUM(CASE WHEN Churn = 'Yes' THEN 1 ELSE 0 END) AS churned,

```

*ROUND(SUM(CASE WHEN Churn = 'Yes' THEN 1 ELSE 0 END) / COUNT()) * 100,*
2) AS churn_rate_pct FROM customers GROUP BY Contract;

SELECT Contract, ROUND(SUM(CASE WHEN Churn = 'Yes' THEN
MonthlyCharges ELSE 0 END) * 12, 2) AS annual_revenue_lost,
ROUND(SUM(CASE WHEN Churn = 'No' THEN MonthlyCharges ELSE 0 END) *
12, 2) AS annual_revenue_retained FROM customers GROUP BY Contract;

Output:

	Contract	annual_revenue_lost	annual_revenue_retained
	One year	169421.40	980377.80
	Month-to-month	1450165.20	1637364.60
	Two year	49983.60	1186086.60